



Paper id: 252172

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Subject Code: BMI203

Roll No:

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MCAINT
(SEM II) THEORY EXAMINATION 2024-25
DATA STRUCTURE

TIME: 3 HRS**M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably

SECTION A**1. Attempt all questions in brief.****02 x 7 = 14**

Q no.	Question	CO	Level
a.	Define stack and mention its applications.	1	K1
b.	Write a postfix expression for the infix expression $(A+B)*C$.	1	K1
c.	Write advantage of using a linked list over an array.	2	K1
d.	Differentiate between binary tree and binary search tree.	3	K1
e.	What is a Huffman tree used for?	3	K1
f.	What is the time complexity of merge sort?	4	K1
g.	What is a spanning tree?	5	K1

SECTION B**2. Attempt any three of the following:****07 x 3 = 21**

Q no.	Question	CO	Level
a.	Define data structures. Explain the classification of data structures with examples.	1	K2
b.	Explain the operations of circular queue using arrays. Simulate the operations: Enqueue(10), Enqueue(20), Dequeue(), Enqueue(30), Enqueue(40), Dequeue() on a circular queue.	2	K3
c.	Construct a binary search tree using: 60, 20, 80, 10, 30, 70, 90. Show inorder traversal.	3	K3
d.	Explain the complexity analysis of quick sort and merge sort. Which one is better for large data and why?	4	K2
e.	What is hashing in the context of file organization? Explain primary and secondary hashing with examples.	5	K2

SECTION C**3. Attempt any one part of the following:****07 x 1 = 07**

Q no.	Question	CO	Level
a.	Explain recursion. Describe Tower of Hanoi problem using recursion. Solve the Tower of Hanoi problem for 3 disks and trace the moves.	1	K2
b.	Convert the infix expression: $(A+B)*(C-D)$ to postfix and evaluate it using $A=2, B=3, C=5, D=1$.	1	K3

4. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Write algorithm and program to create a singly linked list and perform insertion and deletion at the beginning, middle, and end.	2	K2
b.	Construct an expression tree for the postfix expression $5\ 3\ +\ 8\ 2\ -\ * \ 4\ /$ and evaluate the result.	2	K3

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5. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question	CO	Level
a.	Create an AVL tree by inserting the values: 10, 20, 30, 40, 50. Show the rebalancing at each step.	3	K2
b.	Define binary tree. Explain different types of binary trees and their properties.	3	K2

6. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question	CO	Level
a.	What is a heap? How is it used in heap sort? Explain min-heap and max-heap.	4	K2
b.	Sort the following array using Quick Sort: Array: [15, 3, 9, 8, 5, 2, 7, 1, 6] What is its best, worst, and average case performance?	4	K3

7. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question	CO	Level
a.	Represent the graph with vertices A, B, C, D, E and edges A-B, A-C, B-D, C-E as: • Adjacency matrix • Adjacency list	5	K3
b.	Explain depth-first search and breadth-first search. Compare their traversal strategies using suitable example. https://www.aktuonline.com	5	K2